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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Ctrl+S
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 8
Project / Solution Name	Aiversecity
Name (Leader)	Yong Jia Wei
Email & Phone (Leader)	contact@davidyong.dev / 01110576001
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Many students know they should prepare for a changing job market, but they often do not know which career direction fits their current strengths or what to learn next. Career advice is usually generic, scattered across many websites, or focused only on job titles. This can leave students choosing courses, projects, and skills without a clear reason.

The problem is especially relevant to university students and early-career learners who are exploring data, software, and analytical roles. They need practical guidance that connects their self-reported strengths with realistic role requirements and learning steps. Without this, students may spend time on irrelevant skills, underestimate gaps in communication or problem solving, or feel discouraged by broad claims about AI replacing jobs.

Aiversecity addresses this gap by turning career exploration into a short, guided experience. It focuses on three pathways: Data Analyst, Software Developer, and Operations Research Analyst. The aim is not to promise employment outcomes. Instead, it helps learners reflect on their readiness, understand role expectations, and identify useful next steps. This supports more informed learning decisions and contributes to SDG 8: Decent Work and Economic Growth.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Aiversecity is a career-discovery web application for students who are preparing for a future shaped by data and AI. Users complete a short assessment covering AI literacy, data fluency, problem solving, critical thinking, adaptability, and communication. The application compares the responses with transparent role profiles and suggests suitable career pathways, learning priorities, and practical next steps.

The current prototype does not call a GenAI model or make automated employment decisions. Instead, it uses explainable scoring rules so that users can understand why a pathway is recommended. This is important for a student-facing tool: the result is a starting point for exploration, not a prediction of employability.

The prototype also presents dated public occupation and news context where available, while clearly distinguishing evidence from general career advice. It is built and run on Replit, which provides the team's cloud development and deployment environment. For a later Phase 2 version, a carefully scoped GenAI feature could help explain unfamiliar terms or turn a selected pathway into a more personalised learning plan, subject to source checking and clear user controls.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

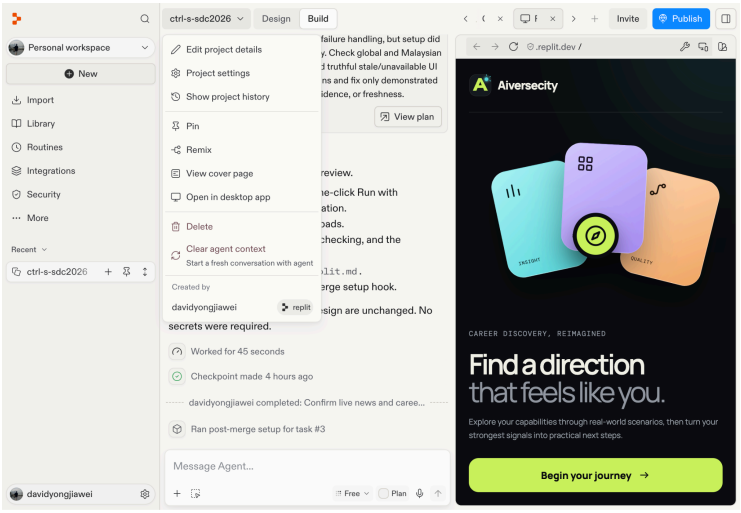
Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

The functional prototype uses React and TypeScript for the interface, Vite for development and builds, and Tailwind CSS for styling. It runs on Replit, where the team imported, developed, and tested the project. The deployed prototype is also available through Replit.

The application is intentionally lightweight. Assessment answers remain in the browser session and are processed using clear, transparent scoring rules. There is no user account system, database, GPU workload, or runtime GenAI model. A small server layer retrieves selected public sources, including dated occupation information and official RSS feeds, when available. If a source cannot be refreshed, the interface should clearly indicate that the evidence is unavailable rather than presenting it as current.

The main user flow is straightforward. A user completes the assessment, the browser applies the scoring rules, and the application presents three career pathway matches along with learning priorities and explanations. Public occupation and news sources are fetched and validated by the server, then shown in the interface with their dates so users can understand the context.

The official sponsor platform used is Replit. It was used to import and run the team project, and a screenshot of the project workspace and preview is included as proof. RunPod was not used because the current prototype has no GPU workload. No sponsor-provided GenAI API was used in the running application.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We started by focusing on a specific type of user, a university student who wants some direction about their career but is unsure how their current strengths connect to different roles. Instead of trying to build a broad job-search platform, we kept the scope manageable by focusing on three clearly different pathways and a short assessment.

To make the experience easy to use, we organised it into a simple flow that moves from the introduction to the assessment, results, pathway details, and next steps. The recommendations also explain why a particular result was given, rather than showing a score without any context. The interface is designed to work smoothly on mobile while still being easy to read on a desktop.

From a technical perspective, the project does not collect real CVs or personal records. It uses fictional demo profiles and responses that only last for the session. Public source data is used to provide context, not to claim that it proves local hiring demand. The app is deployed on Replit and can be reviewed through a public project link before submission.

Before moving on to Phase 2, the team would test the flow with more students, check whether the role weightings feel fair, improve the relevance of the Malaysian labour-market information, and only add an AI feature if it can offer useful and explainable support without pretending to know more than it does.

4. Preliminary Environmental & SDG Impact Framework

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.6: support young people in moving towards employment, education, or training by helping them identify realistic career directions and next learning steps. The project also supports SDG 4.4, which focuses on increasing relevant technical and vocational skills for employment and decent work.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Malaysia's youth unemployment rate for people aged 15–24 was 10.2% in June 2026, representing about 292,200 unemployed young people. This does not mean career guidance alone can solve unemployment; it shows why students need earlier, clearer support when exploring career and skill pathways. For Aiversecity itself, the current baseline is 0 pilot users measured, so no outcome is claimed yet. Source: Department of Statistics Malaysia, OpenDOSM Youth Unemployment dataset.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage of pilot users who complete the Aiversecity assessment and can identify one relevant career pathway plus one specific next learning action in a short exit check.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	In a small Phase 2 pilot with at least 20 voluntary student testers, target 70% or more of completed users being able to state a relevant pathway and one next learning action after using the prototype. This measures career-direction clarity, not employment placement or salary outcomes.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	**Type (code/dataset/model)**	**License**
React and React DOM	Code framework	MIT
TypeScript	Code language/tooling	Apache License 2.0
Vite and React plugin	Build tool	MIT
Tailwind CSS	Styling framework	MIT
fast-xml-parser	Code library	MIT
Playwright and tsx	Testing/development tools	Apache License 2.0 / MIT
BLS Occupational Outlook Handbook information	Public reference data	U.S. public information, attribution provided
O*NET/BLS occupation descriptions	Public reference data	Provider terms and attribution
UN News and MOSTI RSS titles	Public reference data	Provider terms, linked and attributed

Manrope and DM Mono	Font files	SIL Open Font License 1.1

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



25 October 2026